

The Measurement of AI Exposure and Its Impact on Labor Demand in China: Evidence from Large Language Models*

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Abstract

Large language models (LLMs) may displace exactly the high-skill, white-collar work that earlier waves of automation left untouched. We measure this exposure directly for the Chinese labor market. Using 1.25 million online job postings from January 2018 to May 2024 and GPT-4 ratings of the O*NET work activities and tasks each posting maps to, we build an occupation-level LLM exposure index from Chinese posting text itself, avoiding the usual step of matching Chinese occupations to US titles. The average exposure of newly posted jobs fell over the sample period, and the most exposed occupations are knowledge-intensive white-collar jobs—accountants, editors, salespeople, and programmers. Instrumenting Chinese exposure with the exposure of matched US occupations, we find that a one-standard-deviation increase in exposure lowers an occupation’s share of new postings by about 2.3 percentage points and widens within-occupation pay dispersion. The pattern contrasts with US evidence and suggests that China’s labor market has not yet reorganized work around the new technology.

Keywords: artificial intelligence; large language models; labor demand; employment; ChatGPT

JEL: E20, J01, J24, O30

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1 Introduction

Every general-purpose technology since the steam engine has reshaped who works and at what wage. The technologies of the last half century—computers, the internet, industrial robots—mostly automated routine manual and clerical work and left cognitive, non-routine work to skilled labor. Large language models (LLMs) may be different. Because they process natural language and perform rule-based cognitive tasks, they reach into occupations that require education and pay well, raising the prospect that this wave of automation falls on white-collar workers.

The standard way to gauge which jobs are at risk is to build an occupation-level “exposure” index that compares what a technology can do with what each occupation requires. These indices are almost always built on US data. The Occupational Information Network (O*NET), organized around the US Standard Occupational Classification, describes roughly a thousand occupations in terms of their tasks, work activities, and skills, and a large literature has used it to construct exposure measures for successive digital technologies (Frey and Osborne, 2017; Webb, 2020; Felten et al., 2021; Eloundou et al., 2024). Eloundou et al. (2024), for example, estimate that about 80% of US workers have at least 10% of their tasks exposed to GPT models. Work on China typically imports these US-based measures by matching Chinese occupations to US occupation titles—a step that is problematic when the same title covers different tasks in the two countries.

This paper measures LLM exposure for China directly, from the text of Chinese job postings. We draw on 1.25 million postings randomly sampled from Zhaopin.com between January 2018 and May 2024, and we link the detailed work content of each posting to the roughly 2,000 “detailed work activities” (DWAs) and 19,000 tasks that O*NET uses to describe occupations. We ask GPT-4 to rate how exposed each work activity and task is to LLM technology, and we aggregate these ratings—weighted by their importance within each posting—to the occupation level. Because the index is built from each posting’s own description of its work, it does not require matching Chinese and US occupation titles and so sidesteps the cross-country comparability problem. We then estimate how exposure relates to labor demand and to the education, experience, and pay that postings require, at both the occupation and the city level, using the exposure of matched US occupations as an instrument for potential endogeneity.

Four findings stand out. First, the average LLM exposure of newly posted jobs declined over the sample period, with the fall accelerating in 2021–2022 and again in 2024. Sorting occupations into exposure quartiles, demand grew for the least-exposed occupations and shrank for the most-exposed, consistent with a substitution effect concentrated among highly exposed work. Second, the twenty most exposed occupations are knowledge-intensive white-collar jobs—accountants, editors, salespeople, and programmers. Third, exposure predicts weaker labor demand. A one-standard-deviation increase in occupation-level exposure lowers the occupation’s share of new postings by about 2.3 percentage points in our instrumented estimates; against a mean occupational share of 1.4%, this is a large contraction. Highly exposed occupations also show rising within-occupation pay dispersion and, at the city level, significantly higher education and experience requirements. Fourth, tracking the work content of postings within occupations over time, we find little systematic change: the Chinese labor market has not, so far, reorganized tasks in response to the new

technology.

This paper draws on two literatures. The first measures occupational exposure to new technologies. Technological change can complement or substitute for workers (Autor et al., 2022), but the two are hard to separate empirically, so the literature usually stops at an “exposure” measure of a technology’s potential reach; Staneva and Elliott (2023) classify such measures into five families based on tasks, patents, job postings, skill rubrics, and skills. The task-based approach, building on Autor et al. (2003), is the most common: it decomposes an occupation into tasks, assesses substitutability task by task, and aggregates. O*NET is the workhorse source. Frey and Osborne (2017) estimate automation probabilities for 702 occupations; Brynjolfsson and Mitchell (2017) rate tasks for their suitability to machine learning; Webb (2020) matches AI patents to O*NET job descriptions; Felten et al. (2021) link AI application areas to O*NET abilities; and, in the LLM era, Eloundou et al. (2024) and Felten et al. (2023) score occupational exposure to language models. We follow this line—especially Eloundou et al. (2024)—but build the index from Chinese posting text. Two caveats carry through: an exposure index measures capability overlap, not realized displacement, which depends on how firms and workers adapt (Autor et al., 2022); and, unlike the industrial-robot literature that exploits Bartik-style adoption shocks (Chen et al., 2022; Wang and Dong, 2020), AI offers no comparable single shock.

A related strand reads occupational change from text, typically from occupation dictionaries (Autor et al., 2022; Wang and Dong, 2023; Hu et al., 2024); we use posting text instead, because the titles actually used in the labor market need not follow the dictionary and postings describe current work content more promptly.

The second literature estimates how technology affects employment. Since the 1970s, computerization has substituted for middle-skill routine work while complementing high-skill abstract work—the “polarization” documented by Autor et al. (2003), Acemoglu and Autor (2011), Autor and Dorn (2013), and Goos et al. (2014). Industrial robots show sizable displacement (Acemoglu and Restrepo, 2019; Graetz and Michaels, 2018; Bessen, 2019; Anelli et al., 2021), and in China they reduce labor demand and wage income, deter in-migration, and push workers toward non-routine jobs (Wang and Dong, 2020; Li et al., 2021; Yan et al., 2020; Giuntella et al., 2022; Wang et al., 2023).

Evidence on AI is more mixed. Frey and Osborne (2017) put 47% of US jobs at high automation risk, whereas Arntz et al. (2016) argue the true share is closer to 9%. Using online vacancies, Acemoglu et al. (2022) find that more AI-exposed establishments post more AI vacancies yet show no association between AI exposure and overall employment or wages, and Hample et al. (2025) likewise find that AI-related hiring crowds out other postings while leaving aggregate employment and wage effects too small to detect. For LLMs, Eloundou et al. (2024) estimate that about 80% of US workers have some tasks affected. For China, Zhou et al. (2021) find that automation hits vulnerable groups hardest and Wang et al. (2022) that substitution risk rises with the “thinking” content of tasks; at the firm level, AI adoption raises the share of educated and younger employees (Yin et al., 2023), shifts the routine/non-routine labor mix (Chen et al., 2024), and raises productivity through skill upgrading (Yao et al., 2024).

Against this backdrop we make three contributions. First, we construct an AI exposure index from the task content of Chinese postings rather than from a China–US title crosswalk, so the measure reflects

Chinese labor-market conditions more accurately. Second, we provide timely, high-frequency evidence—monthly, through May 2024—on how LLM exposure relates to hiring demand and job requirements since generative AI became widely available. Third, we document how a large developing economy’s labor market is adjusting to LLMs, and the pattern contrasts sharply with the US evidence of Acemoglu et al. (2022) and Humpole et al. (2025): where US demand for AI-related work rose without detectable aggregate harm, relative demand for the most exposed Chinese occupations has fallen.

The rest of the paper proceeds as follows. Section 2 describes the recruitment data and the construction of the exposure index. Section 3 presents the results: how exposure has moved over time and across occupations, how it relates to labor demand at the occupation and city levels, and whether occupations have adjusted from within. Section 4 compares China with the United States and discusses what the findings imply.

2 Recruitment Data and the Exposure Index

2.1 Online recruitment data

Firms’ hiring decisions respond quickly to changes in business models and technology, which makes vacancy data a natural window on technological change. US studies commonly use Burning Glass Technologies (now Lightcast) vacancy data for this purpose (Hershbein and Kahn, 2016; Acemoglu et al., 2022). We use data from Zhaopin.com, an online recruitment platform founded in 1994 that holds more than 30% of China’s online recruitment market, with about 230 million active individual users and roughly six million firm users. Zhaopin data have been used to study employer gender preferences (Kuhn and Shen, 2013) and the effect of the COVID-19 pandemic on hiring and global supply chains (Fang et al., 2020).

Online postings are not representative of the labor market as a whole: they over-represent younger, more educated, and better-paid urban positions (Kuhn and Shen, 2013). Relative to conventional labor-force surveys, platform data are less representative and do not record the final match between workers and firms. But they are timely, high-frequency, and fine-grained, and they capture firms’ hiring demand as it evolves—a leading indicator of how new technology is reshaping demand. For our question, the effect of LLMs on white-collar work, the platform’s over-representation of exactly those positions is an asset rather than a liability.

Our main sample is a nationwide random sample of postings from January 2018 to May 2024. Sampling is stratified by year; after reweighting, the effective sample is about 1.25 million postings (Appendix A) covering 346 occupations grouped into 78 minor groups. Table 1 lists the five most-recruited occupations and their market shares by year. Sales representatives and sales managers dominate the early years; by 2022–2024 “production workers, all other” rises sharply, reaching 17% of postings in early 2024.

Table 1: Five most-recruited occupations and their market share, by year

Year	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
2018	Sales Reps., Wholesale & Mfg. (11.8)	Sales Managers (8.1)	General & Operations Managers (5.0)	Marketing Managers (4.8)	Executive Secretaries & Admin. Assistants (4.2)
2019	Sales Reps., Wholesale & Mfg. (11.8)	Sales Managers (7.8)	Marketing Managers (4.5)	General & Operations Managers (4.3)	Executive Secretaries & Admin. Assistants (3.8)
2020	Sales Reps., Wholesale & Mfg. (11.8)	Sales Managers (7.2)	Marketing Managers (4.3)	Customer Service Reps. (4.2)	Production Workers, All Other (3.7)
2021	Sales Reps., Wholesale & Mfg. (10.5)	Sales Managers (5.4)	Customer Service Reps. (4.2)	Marketing Managers (4.0)	Production Workers, All Other (3.7)
2022	Production Workers, All Other (8.7)	Sales Reps., Wholesale & Mfg. (8.6)	Customer Service Reps. (5.4)	Sales Managers (4.1)	Marketing Managers (3.5)
2023	Production Workers, All Other (11.3)	Sales Reps., Wholesale & Mfg. (7.3)	Sales Managers (4.6)	Customer Service Reps. (4.5)	Human Resources Specialists (4.3)
2024 (Jan–May)	Production Workers, All Other (17.0)	Sales Reps., Wholesale & Mfg. (5.2)	Sales Managers (4.0)	Human Resources Specialists (3.5)	Supply Chain Managers (3.1)

Notes: Market share (%) of all sampled job postings in parentheses. Occupations are labeled by their O*NET titles. Source: sampled Zhaopin job-posting data.

2.2 Constructing the LLM exposure index

Following the literature, and Eloundou et al. (2024) in particular, we use O*NET’s decomposition of work content into two complementary levels—detailed work activities (DWAs) and tasks—to build the index. Figure 1 illustrates these levels. For an electrician, specific tasks such as “prepare sketches or follow blueprints to determine the location of wiring and equipment” are clustered by O*NET into DWAs (here, “create construction or installation diagrams”), and DWAs in turn roll up into broader intermediate and generalized work activities. O*NET collects about 19,000 task statements through surveys and interviews and clusters them into about 2,000 DWAs; the DWA and task levels are the most detailed and informative for distinguishing occupations.

Figure 2 shows the construction strategy. Consider the DWA route. We take O*NET’s 2,087 DWAs as the units of work content and, with GPT-4, score each DWA’s exposure to LLM technology, denoted s_i^d . We then match the text of each posting j to the DWAs it contains, $(d_x, \dots, d_y)$, together with their importance weights within the job, $(w_x, \dots, w_y)$ with $\sum w = 1$. The importance-weighted sum of the corresponding scores, $\sum_k w_k s_k^d$, gives posting j ’s exposure. Averaging over all postings in an occupation (or occupation group) yields the “DWA-based LLM exposure index.” The task route is analogous, using O*NET’s 19,265 tasks, with each posting matched to up to twenty tasks. To keep the GPT-4 scoring tractable, we draw a stratified measurement sample of 77,000 postings (1,000 per month) on which both indices are computed.

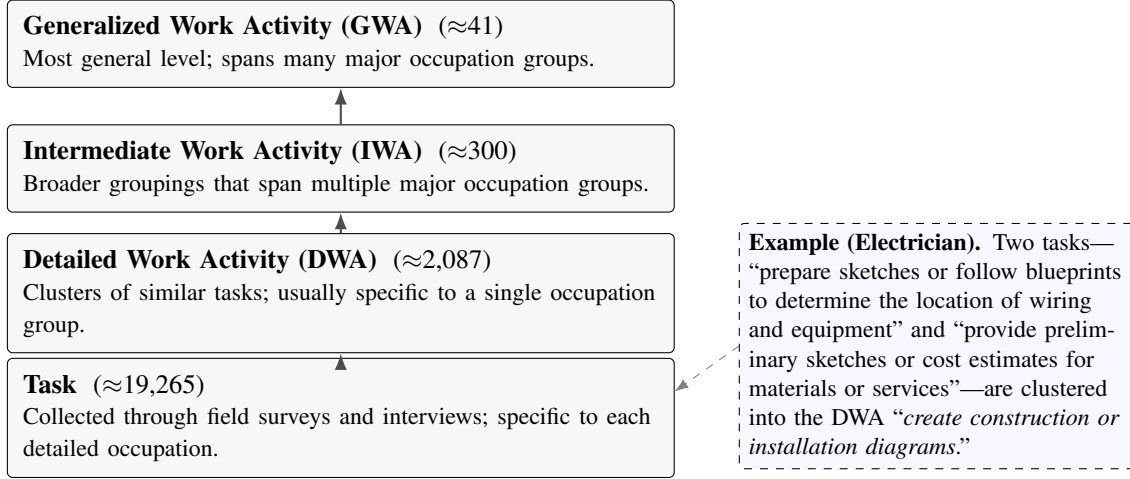


Figure 1: Levels of work-content descriptors in O*NET, from most detailed (Task) to most general (GWA). More detailed levels are more informative for distinguishing occupations. Arrows denote upward aggregation.

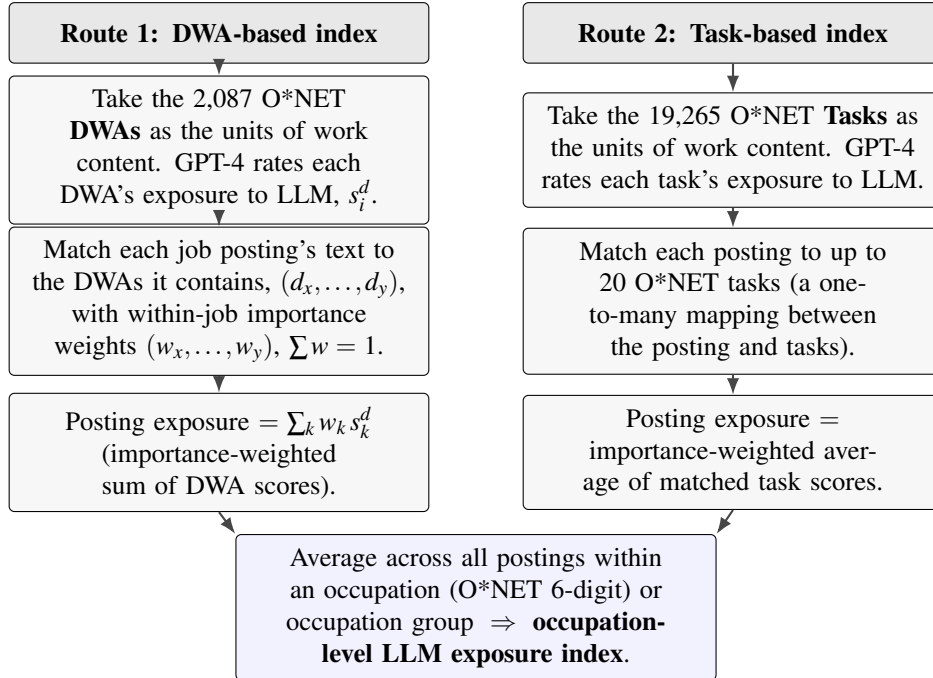


Figure 2: Two complementary strategies for constructing the occupation-level LLM exposure index. Both start from the detailed work-content of each job posting and therefore do not rely on matching Chinese and US occupation *titles*.

Appendix E sets out the structure of this construction and is explicit about which parts of it we have checked and which we have not; Appendix F documents the revised procedure adopted for the public release, together with its prompts.

Both routes start from each posting’s own work content and so do not rely on a correspondence between Chinese and US occupation titles; they provide an independent assessment for each Chinese occupation. For comparison, and to construct an instrument, we also match Chinese occupation titles to US titles and merge in the LLM exposure index of Felten et al. (2023), forming a title-based “Felten–LLM” index. The Felten–LLM index correlates with our DWA-based and task-based indices at 0.75 and 0.71 respectively, and the two original indices correlate at 0.89 (Appendix B, Figure A1). We report DWA-based results in the main text; the task-based results, which are very similar, are in Appendix C.

3 Exposure and the Labor Market

3.1 Exposure over time

To see how demand for exposed work has evolved, we compute, for each month, the weighted-average exposure of newly posted jobs, standardized and seasonally adjusted. Figure 3 plots the three indices. Over more than six years, the average exposure of new postings falls steadily, with the decline accelerating from early 2021 through late 2022 and again in 2024. New postings are increasingly concentrated in low-exposure work, consistent with LLM technology substituting for more exposed labor.

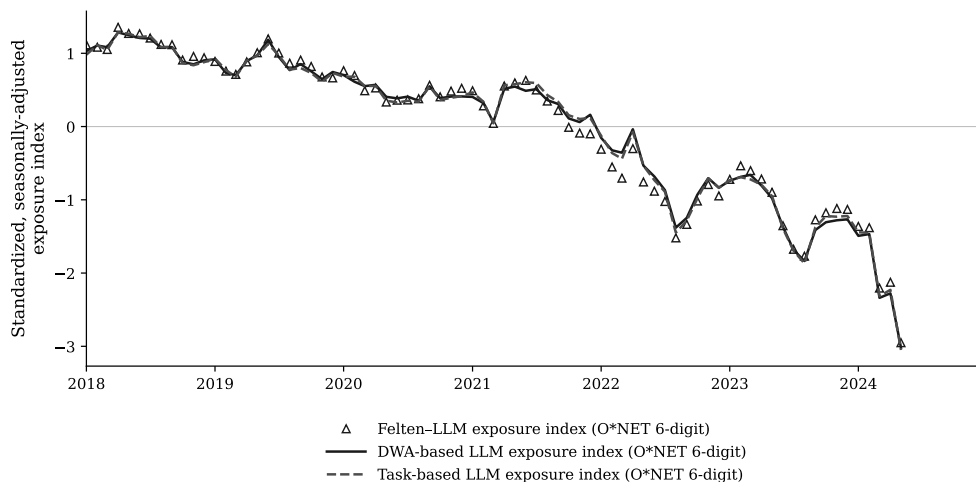


Figure 3: Monthly trend in the LLM exposure of newly posted jobs. Each series is the importance-weighted average exposure of new postings, standardized and seasonally adjusted. “Felten–LLM” is the title-based index; “DWA-based” and “Task-based” are built from posting text. The DWA- and task-based series nearly coincide.

Figure 4 sorts occupations into four groups by DWA-based exposure and tracks each group’s share of new postings. The least-exposed occupations gain share, while the more-exposed groups lose share, most steeply in the top group. The aggregate decline in Figure 3 is thus driven by rising demand for low-exposure work alongside falling demand for high-exposure work.

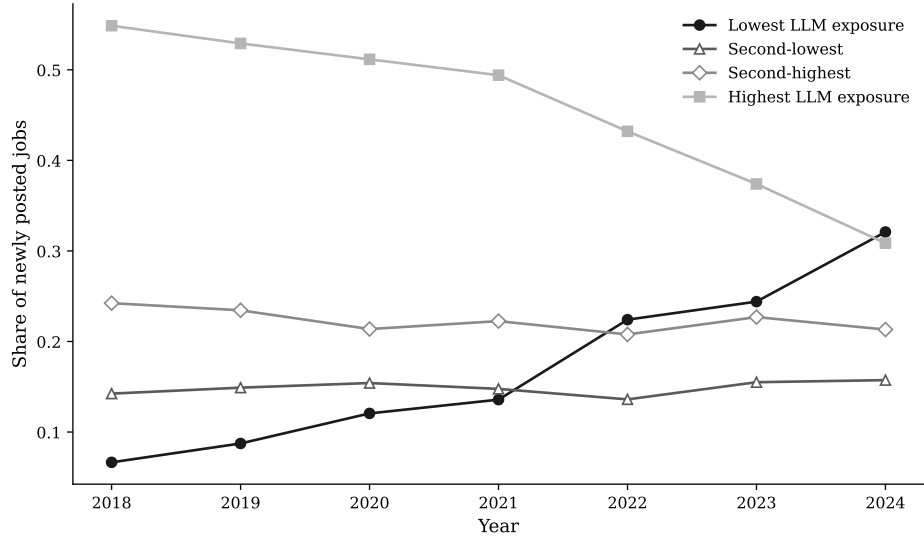


Figure 4: Share of newly posted jobs over time, by exposure quartile (DWA-based index). Quartiles run from the least-exposed to the most-exposed occupations.

3.2 Which occupations are exposed

Figure 5 lists the twenty occupations with the highest and lowest DWA-based exposure. The most exposed are knowledge-intensive white-collar jobs—accountants, editors, salespeople, and programmers—whose work involves extensive text processing and information integration, with tasks that are largely rule-based and repetitive and make limited demands on creativity. Sales occupations score high, which may seem surprising given the “soft skills” the work requires; but many sales tasks—collecting customer information, phone outreach, answering queries—are routine and hence exposed. Because LLMs are a general-purpose technology with strong natural-language and rule-execution abilities, they reach more occupations, and more white-collar ones, than earlier digital technologies. The least-exposed occupations—food-service workers, production workers, and domestic-service workers—involve manual work and direct interpersonal interaction; these jobs are little affected by LLMs, though they may be exposed to robots and other technologies. Appendix D describes one firm’s experience automating customer-service work.

Table 2 relates exposure to three occupational characteristics across 72 minor groups, using OLS with the standardized DWA-based index as the dependent variable. Occupations whose postings require more education (column 1) and pay more (column 3) are more exposed. In the multivariate specification (column 4), education and pay retain positive coefficients while required experience turns negative—conditional on education and pay, more exposed occupations require somewhat less experience. That exposure is concentrated in high-education, high-pay occupations echoes Eloundou et al. (2024) for the United States.

3.3 Exposure and demand for new hires

Figure 6 plots the change in an occupation’s share of new postings—the most recent 24 months (June 2022–May 2024) relative to 2019–2020—against its exposure. The relationship is negative: more exposed

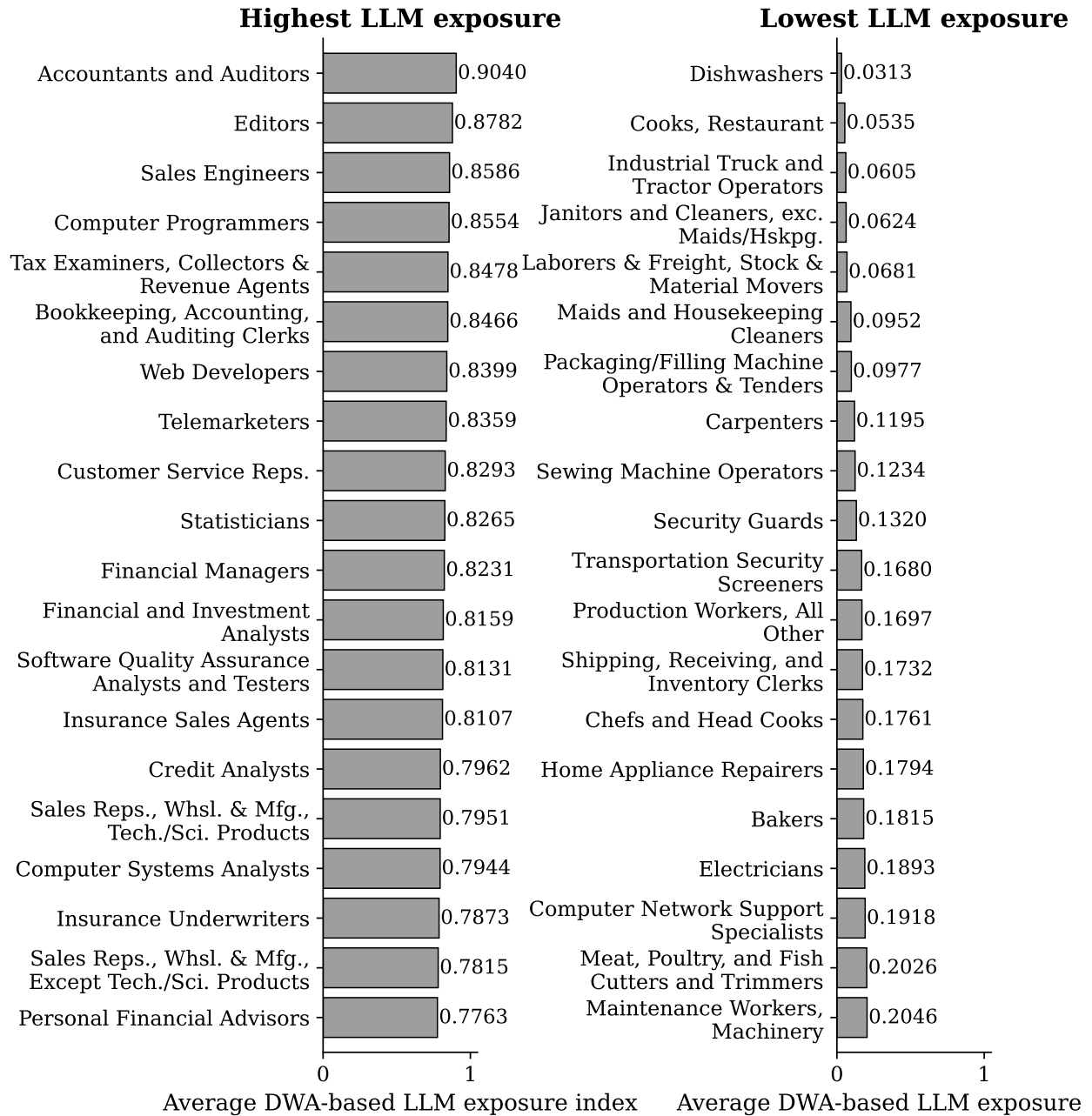


Figure 5: Occupations with the highest and lowest DWA-based LLM exposure. Bars show the average exposure index; occupations are labeled by O*NET title.

Table 2: DWA-based LLM exposure and job-posting occupational characteristics

	(1)	(2)	(3)	(4)
<i>Dependent variable: standardized DWA-based LLM exposure index</i>				
Required education (years)	0.150*** (0.0180)			0.173*** (0.0247)
Required experience (years)		0.354** (0.175)		−0.586*** (0.197)
Salary (RMB thousand)			0.193*** (0.0621)	0.123** (0.0518)
Observations	72	72	72	72
R^2	0.438	0.042	0.147	0.528

Notes: Bootstrap standard errors (1,000 replications) in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels. The three regressors are averages computed over postings from January 1, 2018 to May 31, 2024 at the level of 72 occupation minor groups.

occupations gained fewer (or lost more) postings. The fitted slope implies that a one-standard-deviation increase in exposure—roughly the gap between “engineers” and “information and record clerks”—lowers an occupation’s share of new postings by about 1.9 percentage points (significant at 1%). Given a mean occupational share of 1.4%, this is a steep decline. Dropping “other production occupations,” whose share moved by more than five percentage points, leaves the negative relationship intact (right panel).

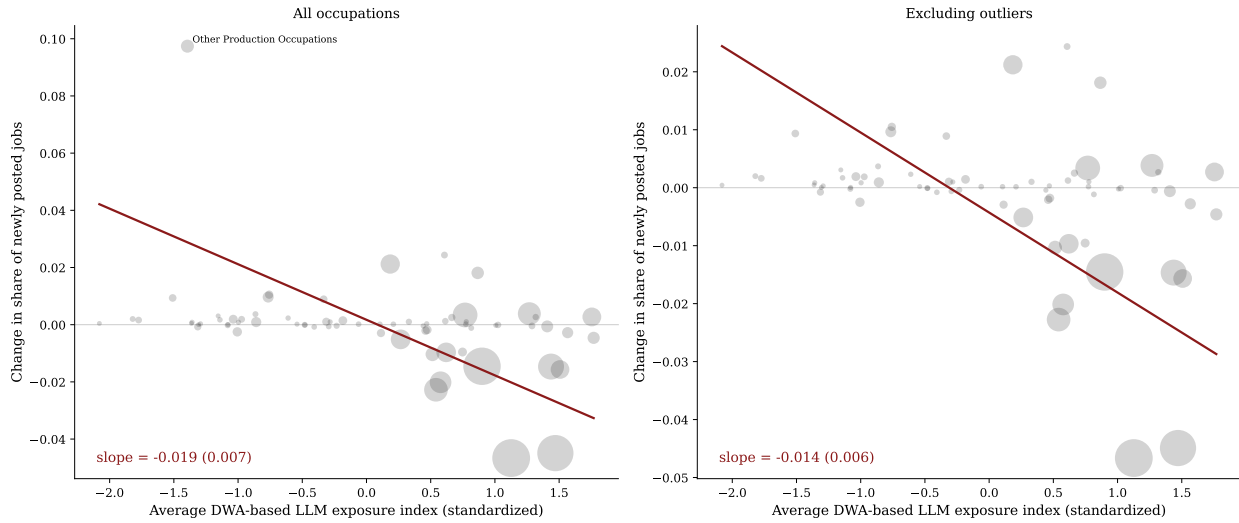


Figure 6: Change in the share of newly posted jobs against DWA-based exposure, by occupation. Each circle is an occupation minor group, sized by 2018 posting volume; the line is a volume-weighted *descriptive* fit (a change of 0.01 on the vertical axis is one percentage point). The right panel excludes the outlying “other production occupations” and is drawn on a narrower vertical scale.

To move toward a causal reading, we instrument Chinese occupation exposure with the Felten–LLM exposure of the matched US occupation. As an early adopter with a mature LLM ecosystem, the United States provides occupation-level exposure that reflects the global technological frontier and is strongly correlated with the shock to the corresponding Chinese occupation, yet US occupational exposure plausibly affects the

Chinese labor market only through Chinese exposure—the exclusion restriction we invoke. We estimate

$$\Delta Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad X_i = \gamma_0 + \gamma_1 \text{Felten}_i + u_i, \quad (1)$$

where ΔY_i is the change, over the most recent 24 months relative to 2019–2020, in occupation i 's share of postings, average required education, average required experience, average salary, or the coefficient of variation of salary; X_i is the standardized (DWA- or task-based) exposure index; and Felten_i is the instrument.

Table 3 reports the occupation-group results. The first stage is strong ($F = 104.5$). In the instrumented estimates, a one-standard-deviation increase in exposure lowers an occupation's share of postings by 2.3 percentage points. We find no significant effect on the average salary an occupation offers, but the coefficient of variation of salary rises, indicating that pay dispersion *within* exposed occupations has widened. Required education and experience also rise with exposure, though the coefficients are not significant in the instrumented specification.

Table 3: DWA-based LLM exposure and changes in labor demand: occupation-group regressions

	(1) First stage	(2) Δ Market share (%)	(3) Δ Educ. (years)	(4) Δ Exper. (years)	(5) Δ Salary (RMB)	(6) Δ CV of salary
<i>Panel A. OLS</i>						
DWA-based LLM exposure		−0.0197*** (0.00695)	0.596*** (0.146)	0.0886** (0.0359)	284.5 (189.2)	0.0570** (0.0260)
<i>Panel B. Instrumental variables</i>						
Felten–LLM exposure (first stage)	0.845*** (0.0827)					
DWA-based LLM exposure		−0.0234*** (0.00467)	0.324 (0.198)	0.0400 (0.0393)	27.74 (227.0)	0.101*** (0.0293)
First-stage F	104.5					
Observations	70	70	70	70	70	70

Notes: OLS regressions report robust standard errors; IV regressions use small-sample-adjusted standard errors (in parentheses). All regressions are weighted by each occupation's 2018 market share. ***, **, * denote significance at the 1%, 5%, and 10% levels. Column (1) is the first stage, instrumenting the DWA-based exposure index with the Felten–LLM exposure index of the matched US occupation. In columns (2)–(6) the dependent variable is the change between June 2022–May 2024 (the most recent 24 months) and 2019–2020 in, respectively, the occupation's share of postings, average required education, average required experience, average salary, and the coefficient of variation of salary.

We then move to cities, treating each as a local labor market. We recompute exposure at the city level as

$$X_c = \sum_{i=1}^I v_{ci} X_i, \quad (2)$$

where v_{ci} is occupation i 's share of city c 's postings, and we instrument it with a Bartik-style predicted exposure that combines the national (US-based) occupational shock with each city's pre-AI occupational mix:

$$\text{Predict}_c = \sum_{k=1}^K w_{ck} \text{Felten}_k, \quad (3)$$

where w_{ck} is the employment share of occupation k in city c from the 2015 one-percent population sample. Because the weights predate the large-scale development of AI, the predicted exposure is highly correlated with current city exposure but plausibly unrelated to a city’s occupational demand shifts over 2019–2024 (Borusyak et al., 2022). Table 4 reports the city-level results, which point the same way: a one-standard-deviation increase in a city’s exposure is associated with a 1.2-percentage-point contraction in labor demand, greater pay dispersion, and significantly higher education and experience requirements for new postings.

Table 4: DWA-based LLM exposure and changes in labor demand: city-level regressions

	(1) First stage	(2) Δ Market share (%)	(3) Δ Educ. (years)	(4) Δ Exper. (years)	(5) Δ Salary (RMB)	(6) Δ CV of salary
<i>Panel A. OLS</i>						
DWA-based LLM exposure		−0.00335*** (0.00111)	0.604** (0.246)	0.0691** (0.0347)	−67.33 (175.7)	0.0533* (0.0278)
City controls		Y	Y	Y	Y	Y
<i>Panel B. Instrumental variables</i>						
Felten–LLM exposure (first stage)	1.384*** (0.388)					
DWA-based LLM exposure		−0.0119*** (0.00382)	3.000*** (0.948)	0.389*** (0.127)	−741.1* (411.4)	0.212** (0.0989)
City controls		Y	Y	Y	Y	Y
First-stage F	12.74					
Observations	95	95	95	95	95	95

Notes: OLS regressions report robust standard errors; IV regressions use small-sample-adjusted standard errors (in parentheses). All regressions are weighted by each occupation’s 2018 market share. ***, **, * denote significance at the 1%, 5%, and 10% levels. All regressions control for the following city-level covariates (2018 values): GDP per capita, employment shares of the primary and secondary sectors, the city’s share of national postings, and the city’s average required education, required experience, and salary; GDP per capita and sectoral shares come from city statistical yearbooks and the remaining variables from the Zhaopin sample. Column (1) is the first stage, instrumenting the city-level exposure index with a Bartik-style predicted exposure built from US (Felten–LLM) occupational exposure and the city’s 2015 occupational employment structure. The sample comprises the 95 cities whose 2018 postings exceed 0.1% of the national market. Dependent variables in columns (2)–(6) are defined as in Table 3.

3.4 Within-occupation adjustment

The analysis so far treats an occupation’s exposure as fixed over time. If instead the work content of an occupation shifts—and shifts in a way correlated with both its exposure and its demand—our estimates could suffer from omitted-variable bias. To check, we relax the assumption and use the 77,000-posting measurement sample to score each posting directly and average by minor group and month. Figure 7 plots the resulting within-occupation exposure trend for the sixteen most common minor groups, ordered by average exposure.

For most occupations the exposure of postings is flat, so these occupations have not visibly restructured their tasks around the new technology. A few groups—information and record clerks, arts and design workers, and operations managers—show a downward drift in exposure over the last two years, hinting that they may be steering work away from tasks that LLMs can do. On the whole, though, the Chinese labor market

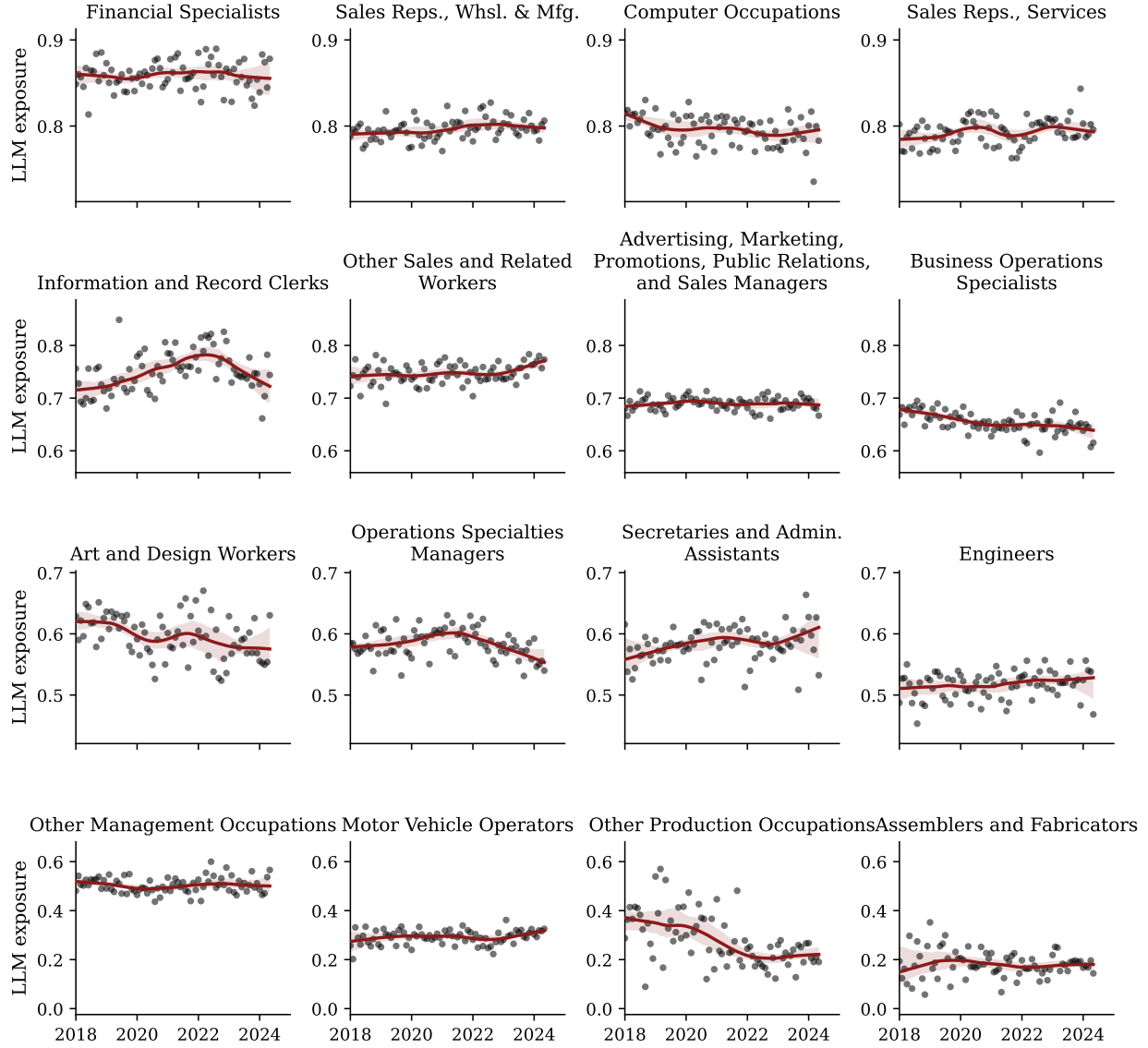


Figure 7: Within-occupation trend in DWA-based exposure for the sixteen most common minor groups. Points are monthly averages from the measurement sample; the line is a nonparametric smoother with a confidence band.

shows little systematic within-occupation adjustment during the sample period.

4 Discussion

From computers and the internet to industrial robots and now AI, digital technologies have repeatedly reorganized production and reshaped the labor market. Generative AI differs from earlier waves in reaching white-collar and higher-paid work. Building an exposure index tailored to Chinese occupations, we document that the demand for exposed work has been contracting and that the labor market has not yet adjusted from within.

The Chinese pattern contrasts with recent US evidence. Acemoglu et al. (2022) find that US demand

for AI-related work rose steadily from 2010, accelerated in 2015–2016, and grew fastest for the most AI-exposed occupations; at the establishment level, a one-standard-deviation increase in AI exposure—about the gap between finance and mining—raised the AI share of vacancies by fifteen percentage points. Yet they detect no association between AI exposure and overall employment or wages, and conclude that, outside AI-specific roles, the technology’s complementarity and productivity gains had so far outweighed displacement. Our results run the other way: in China, the most exposed occupations saw relative demand fall, faced higher education and experience requirements, and experienced wider within-occupation pay gaps.

Several factors could account for the divergence. The two studies cover different periods and technologies—the sample in Acemoglu et al. (2022) predates the LLM era, while ours runs through the diffusion of generative AI. Macroeconomic conditions and other fundamentals also differ. But the contrast is consistent with a simpler reading: the productivity gains from AI depend on how fully the technology is integrated into production, and in the early phase of adoption a labor market that has not yet reorganized work may register substitution before complementarity. Our finding that within-occupation task content has barely moved fits this interpretation. Whether the Chinese pattern reflects a transient adjustment lag or a more durable difference is a question worth tracking as the technology matures; the mechanisms behind the cross-country gap—adoption intensity, the organization of work, and the availability of complementary infrastructure and skills—are natural targets for future research.

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Appendix

A Sampling

Zhaopin.com provided a sample of postings covering January 1, 2018 to May 31, 2024. Sampling is stratified by year, drawing about 200,000 postings per year. Because 2024 contributes only five months, its raw sample also totals about 200,000; to keep annual volumes comparable, each 2024 observation receives a weight of 0.39, the share of a typical year’s postings that appear before May 31 (based on 2018–2023). The weighted sample sums to about 1.25 million. Under this design, posting volumes are comparable across months within a year but not across years.

B Correlation among exposure indices

Figure A1 plots the pairwise relationships among the three exposure indices across the 247 O*NET six-digit occupations in the measurement sample. All three are strongly correlated; the two indices built from Chinese posting data (DWA- and task-based) correlate especially strongly, supporting the two construction routes. Both original indices are also strongly and positively correlated with the US-based Felten–LLM index once matched at the occupation level—the same relationship that underlies the strong first stage of the instrumental-variables estimates. The measurement sample covers 247 of the 346 occupations in the full sample; the remaining occupations are too small to be included, and the 247 covered occupations account for 99.05% of all postings.

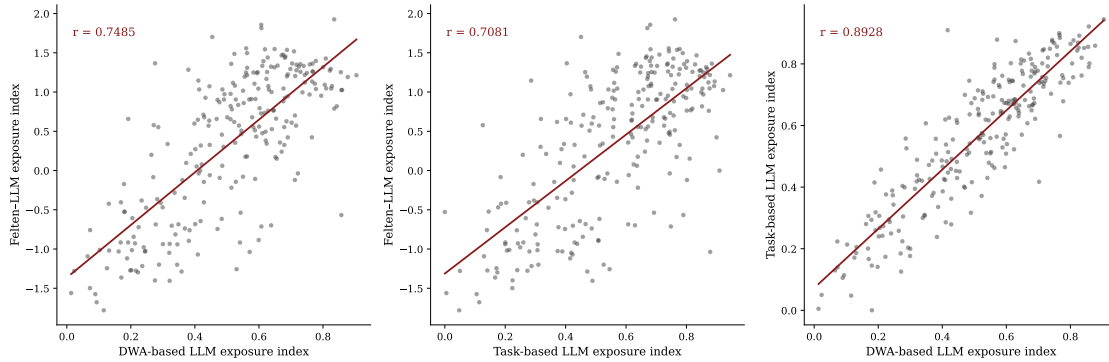


Figure A1: Correlations among the DWA-based, task-based, and Felten–LLM exposure indices at the O*NET six-digit occupation level.

C Task-based results

Because the DWA- and task-based indices are highly correlated, the main text reports only the DWA-based results. Tables A1–A3 report the task-based counterparts of Tables 2–4. The results are very close to the DWA-based estimates: exposure is concentrated in high-education, high-pay occupations, and more exposed occupations and cities experience relative demand contractions and wider within-occupation pay dispersion.

Table A1: Task-based LLM exposure and job-posting occupational characteristics

	(1)	(2)	(3)	(4)
<i>Dependent variable: standardized Task-based LLM exposure index</i>				
Required education (years)	0.153*** (0.0191)			0.153*** (0.0242)
Required experience (years)		0.580*** (0.170)		−0.246 (0.184)
Salary (RMB thousand)			0.203*** (0.0621)	0.104** (0.0506)
Observations	72	72	72	72
R^2	0.455	0.114	0.162	0.491

Notes: Bootstrap standard errors (1,000 replications) in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels. Regressors are averages over postings from January 1, 2018 to May 31, 2024 at the level of 72 occupation minor groups. This is the Task-based counterpart of Table 2.

Table A2: Task-based LLM exposure and changes in labor demand: occupation-group regressions

	(1) First stage	(2) Δ Market share (%)	(3) Δ Educ. (years)	(4) Δ Exper. (years)	(5) Δ Salary (RMB)	(6) Δ CV of salary
<i>Panel A. OLS</i>						
Task-based LLM exposure		−0.0198*** (0.00681)	0.433*** (0.130)	0.0720** (0.0319)	258.4 (181.7)	0.0627** (0.0256)
<i>Panel B. Instrumental variables</i>						
Felten–LLM exposure (first stage)	0.912*** (0.0663)					
Task-based LLM exposure		−0.0217*** (0.00434)	0.301 (0.190)	0.0371 (0.0370)	25.72 (210.7)	0.0940*** (0.0266)
First-stage F	188.7					
Observations	70	70	70	70	70	70

Notes: See Table 3. This is the Task-based counterpart. OLS regressions report robust standard errors; IV standard errors are small-sample adjusted. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Table A3: Task-based LLM exposure and changes in labor demand: city-level regressions

	(1) First stage	(2) Δ Market share (%)	(3) Δ Educ. (years)	(4) Δ Exper. (years)	(5) Δ Salary (RMB)	(6) Δ CV of salary
<i>Panel A. OLS</i>						
Task-based LLM exposure		−0.00446*** (0.00136)	0.768*** (0.235)	0.0943*** (0.0345)	−90.29 (187.9)	0.0601** (0.0268)
City controls		Y	Y	Y	Y	Y
<i>Panel B. Instrumental variables</i>						
Felten–LLM exposure (first stage)	1.388*** (0.389)					
Task-based LLM exposure		−0.0118*** (0.00348)	2.990*** (0.897)	0.388*** (0.119)	−738.8* (406.4)	0.212** (0.0973)
City controls		Y	Y	Y	Y	Y
First-stage F	12.71					
Observations	95	95	95	95	95	95

Notes: See Table 4. This is the Task-based counterpart, with the same city-level controls and the 95-city sample. ***, **, * denote significance at the 1%, 5%, and 10% levels.

D Case study: AI reshaping customer-service work

To illustrate how exposure translates into workplace change, we studied a leading domestic online-services firm that had adopted AI in its customer-service operations well before generative AI became prominent at the end of 2023. Adoption came in two phases. From 2017 to 2021 the firm focused on speech recognition, automated quality inspection, and recommendation systems, raising speech-transcription accuracy to 94% and automating ticket classification and multilingual translation. From 2021 to 2022 it extended AI to multilingual support and self-service, raising the online self-service rate from below 50% to 75%.

AI now handles repetitive tasks such as speech transcription and ticket generation, and self-service covers a substantial share of contacts (35% by phone, 75% online). Human agents have shifted to complex cases—urgent coordination, emotional support, and high-net-worth clients—and the required skills have moved from operational tasks toward communication and problem-solving. After full deployment, the customer-service team shrank from more than 20,000 to about 10,000, with a marked fall in labor costs. Even so, the remaining team is stable and well paid (average tenure of six years, 2% monthly attrition, and pay above the 75th percentile in first-tier cities), as the firm used human–machine collaboration to reorganize the function from a cost center toward a value center.

E Measurement note

An index built with a language model is easy to state in a sentence and hard to audit. This appendix sets out the structure of the construction and is explicit about which parts of it we have checked and which we have not. It is deliberately brief on implementation. The revised pipeline adopted for the public release is the one we document in full, with its prompts, in Appendix F and in the accompanying technical documentation (AI and Economics Laboratory, National School of Development, Peking University and Zhaopin, 2025).

E.1 Two distinct uses of the model

It is worth separating two things the model does here, because they fail in different ways and carry different costs.

The first is *scoring*: rating each unit of the O*NET inventory—all 2,087 DWAs and 19,265 tasks—for its exposure to LLM technology. This is the substantive measurement step. It is done once, and its output is a fixed table reused for every posting, so its cost does not grow with the sample and it can afford more care per unit than anything done posting by posting. Errors here shift the level of the index and, if correlated with occupation characteristics, bias the coefficients that follow.

The second is *matching and weighting*: deciding which O*NET units a given posting contains and how important each one is within that job. This runs once per posting and is what binds the scale of the exercise. Errors here behave more like classical measurement error in the posting-level index: they attenuate the occupation-level average toward its mean rather than shifting it systematically, which is one reason we instrument in Section 3.

E.2 Outline of the construction

The construction proceeds in four blocks.

Score the inventory. Every O*NET DWA and task is rated once for its exposure to LLM technology using the four-level rubric of Eloundou et al. (2024), which defines exposure as a halving of completion time at equivalent quality and instructs the rater to default to the null category under uncertainty—a convention that makes the resulting index conservative. The rubric is reproduced in Appendix F. The result is a fixed table of exposure scores, reused unchanged for every posting.

Narrow the candidate set. Assessing a posting against all 2,087 DWAs, or all 19,265 tasks, is not feasible at scale. Each posting is therefore first associated with a small set of plausible O*NET occupations, and only the work content belonging to those occupations is carried forward.

Judge presence. Each candidate unit is then assessed for whether the posting actually describes it, on a graded scale that separates work the posting states explicitly from work merely typical of the occupation. Only the stronger categories survive. This is what keeps the index anchored to each posting’s own text rather than to its occupation label.

Weight and aggregate. Up to twenty surviving units receive importance weights summing to one, which yields posting-level exposure as defined in Section 2; posting-level values are then averaged to the occupation and city levels.

The two middle blocks are the ones that run once per posting. Throughout, model calls alternate with deterministic joins and parsing, and each step writes its output before the next begins, so intermediate results can be inspected and any step re-run without repeating the work upstream of it. Appendix F describes how the two middle blocks were later restructured for the public release, replacing part of the model’s judgment with a matching step that anyone holding the same two text corpora can recompute.

E.3 What is validated, and what is not

Checks the paper runs. Three, none of which requires trusting the model.

Internal consistency. The DWA and task routes share the same rubric but use different inventories, different granularity, and different matching decisions. They correlate at 0.89 at the occupation level (Appendix B), and every substantive result in the main text is reproduced with the task-based index in Appendix C. Two constructions this different agreeing this closely is evidence that the measure is not an artifact of one particular matching step.

External benchmark. Our indices correlate at 0.75 (DWA) and 0.71 (task) with the title-based Felten–LLM index built independently from US data (Felten et al., 2023). The correlation is high enough to indicate that the measures capture a common construct, and low enough to justify building the Chinese index from posting text rather than importing the US one—which is the paper’s methodological claim.

Coverage. The measurement sample covers 247 of 346 occupations, accounting for 99.05% of postings; the excluded occupations are too thin to score reliably.

A check we inherit rather than run. Eloundou et al. (2024) compared human and model annotations under the same rubric and found them close. Our scoring step rests on that finding. We did not re-run a human annotation exercise on the Chinese sample, and readers who doubt the rubric’s transferability should discount the index accordingly.

Checks we did not run. Three gaps are worth naming, since each is addressable and none is addressed here.

First, the per-posting steps—judging presence and assigning weights—were never benchmarked against human coders on a sample of postings. We know the two routes agree with each other; we do not know how either compares with a careful human reading of the same posting. Second, we did not measure run-to-run stability: repeated calls need not return identical output, and we did not quantify the residual variation or its effect on the occupation-level index. Third, and most fundamentally, no step of the construction observes how long any task actually takes, with or without an LLM. The rubric asks for a judgment about time saved; nothing in the data verifies it.

Limits of the object being measured. Beyond the pipeline, three limitations apply to any index of this kind. Exposure is capability overlap, not adoption: it says what an LLM could plausibly do, not what anyone did (Section 4). Posting text describes advertised work, which need not be performed work, and firms may revise their advertising faster or slower than they revise their jobs. And the platform over-represents urban, educated, better-paid positions (Section 2).

F Public data release and post-publication updates

After the original study, the research team—the AI and Economics Laboratory at the National School of Development, Peking University, together with Zhaopin—released a public version of the exposure index and the underlying occupation-level data on September 19, 2025.¹ The release keeps the core idea of this paper but adopts a more transparent and reproducible measurement procedure, set out in the accompanying technical documentation (AI and Economics Laboratory, National School of Development, Peking University and Zhaopin, 2025); that documentation credits the refined construction to Yu et al. (2026), where it is first detailed and applied.

This appendix documents the released version at the level of detail a user of the public data needs: why the measurement procedure was revised, what the revised procedure does step by step, the prompts it uses, and what the released files contain. Everything reported here is drawn from the public technical documentation or, for the exposure rubric, from Eloundou et al. (2024). The estimates in the main text are those of the published paper and are not affected by the revision; readers interested only in those can stop at Appendix E.

¹Landing page and download request: <https://nsd.pku.edu.cn/xzyj/kyfb/zsfb/ai/542352.htm>; contact aiecon@nsd.pku.edu.cn.

F.1 Why the measurement procedure was revised

The paper extracts standardized tasks from each posting by prompting a model directly. That is workable, but one call decides two things at once: what work the posting describes, and how that work maps onto a standard inventory. Because the two judgments are entangled, a poor match cannot be traced to its cause, and the procedure depends on prompt tuning that is hard to audit at intermediate steps.

The public release separates the two, following Hampole et al. (2025). Extraction is broken into four steps that each leave inspectable output, and the final mapping is handed to a text-similarity comparison that any user holding the same two corpora can recompute. The revision improves interpretability and replicability without changing the conceptual construction of the index.

F.2 The four-step extraction pipeline

1. *Isolate task-relevant text.* Postings mix duties with qualifications, benefits, company description, and boilerplate. This step keeps only the duties and requirements, verbatim, and discards the rest.
2. *Distill specific tasks.* The retained text is rewritten as discrete task statements in an O*NET-comparable form and translated into English, preserving the original meaning and detail. This is what makes the later similarity comparison meaningful: it puts both sides into the same idiom and register.
3. *Screen.* Vague or non-task statements are removed—items too broad to identify any activity (“complete other tasks assigned by the leader”), items that are requirements misfiled as duties, and residual untranslated text. Roughly 10% of extracted items are dropped here. These categories define what the index does *not* count as work.
4. *Match.* Each surviving statement is encoded with Sentence-BERT (768-dimensional, paraphrase-MPNet-base-v2), as is each O*NET task, and assigned to the O*NET task with the highest cosine similarity. Mean match similarity is about 0.724. Unlike the corresponding step in the published version, this involves no model judgment: given the two corpora and the stated encoder, it is deterministic and independently recomputable.

Each of the three model-based steps uses a fixed five-part prompt—task description, detailed instructions, procedural guidance, output template, and a worked example. The structure is not decorative: it is what makes the output regular enough to parse and to compare across runs.

F.3 Prompts

The prompts below are those reproduced in the public technical documentation (AI and Economics Laboratory, National School of Development, Peking University and Zhaopin, 2025). Chinese-language worked examples are replaced here by an English gloss of what they demonstrate.

Exposure rubric. Scoring the O*NET inventory uses the rubric of Eloundou et al. (2024), reproduced in the release documentation from their supplementary materials. It is given below in abridged form; the enumerated examples under each label are in their paper. The same rubric is applied to DWAs, with “task” read as “activity”.

Consider the most powerful OpenAI large language model (LLM). This model can complete many tasks that can be formulated as having text input and text output where the context for the input can be captured in 2000 words. The model also cannot draw up-to-date facts (those from <1 year ago) unless they are captured in the input.

Assume you are a worker with an average level of expertise in your role trying to complete the given task. You have access to the LLM as well as any other existing software or computer hardware tools mentioned in the task. [...] You do not have access to any other physical tools or materials.

Please label the given task according to the taxonomy below.

E0 - No exposure

Label tasks E0 if direct access to the LLM through an interface like ChatGPT or the OpenAI playground cannot reduce the time it takes to complete this task with equivalent quality by half or more. If a task requires a high degree of human interaction (for example, in person demonstrations) then it should be classified as E0.

E1 - Direct exposure

Label tasks E1 if direct access to the LLM through an interface like ChatGPT or the OpenAI playground alone can reduce the time it takes to complete the task with equivalent quality by at least half. This includes tasks that can be reduced to: writing and transforming text and code according to complex instructions; providing edits to existing text or code following specifications; translating text between languages; summarizing medium-length documents; answering questions about a document; [... further enumerated examples ...]

E2 - Exposure by LLM-powered applications

Label tasks E2 if having access to the LLM alone may not reduce the time by at least half, but it is easy to imagine additional software developed on top of the LLM that would. [... enumerated examples ...]

E3 - Exposure given image capabilities

[... as E2, additionally assuming a system that can view, caption, and create images ...]

Step 1: isolate task-relevant text.

[System] Your task is to review a job posting from an online recruitment platform and extract the job tasks and requirements directly from the job posting while ensuring accuracy and structure. Please use the following instructions, recommended steps, samples and output format to respond to user inputs.

[User] Instructions:

- (1) The job posting may describe multiple aspects of the position, but you should focus only on job tasks (the activities and responsibilities the employee will be expected to perform) and requirements (the education, age, experience, qualifications, skills, or other personal characteristics the employer expects the candidate to have).
- (2) Extract only the raw text corresponding to "job tasks" and "requirements." Ignore all other content, such as salary, benefits, company descriptions, and work arrangements.
- (3) Classify correctly. Do not rely solely on section titles. Some sections labeled "Job Responsibilities" may actually describe required skills. [...] If a sentence describes actions performed on the job, it belongs in job tasks. If a sentence describes skills, qualifications,

- experience, age, gender, or other personal attributes, it belongs in the requirements.
- (4) Pay attention to the negative semantics. Sometimes a recruitment advertisement states "There is no need to do a certain task." In such cases, do not categorize this statement under either heading.
 - (5) Ignore promotional or motivational content (e.g., "This could change your life!").
 - (6) Maintain original phrasing. Copy the relevant text verbatim. Do not summarize or paraphrase.
 - (7) Handle ambiguous cases: [...] some job tasks can be derived from a requirement statement. For example, from "Be proficient in using relevant architectural design software such as AUTO CAD, PHOTOSHOP, and 3D MAX", we know the job includes the task "Use architectural design software such as AUTO CAD, PHOTOSHOP, and 3D MAX". In this case, classify the statement as both.
 - (8) Avoid duplication.

A three-step process is suggested: First, filter out the original job posting and only keep information about job tasks and job requirements. Second, reread the original job posting to ensure that no tasks or requirements are neglected in your answer. Third, check the job tasks and requirements in your answer are correctly classified based on the instructions.

Output format (Do not include anything else in the answer. If no relevant information is found, return "No relevant information found" under the respective section): Job Tasks:{1.Task1;2.Task2;...}| Requirements:{1.Requirements1;2.Requirements2;...}.

For reference, here is an example of correctly applied filters: <worked example on a Chinese-language posting for a sales-administration assistant, showing four duties retained, six requirements retained, and the benefits block discarded>

Here is the job posting you need to filter: <job post content>

Step 2: distill O*NET-comparable tasks.

[System] Your task is to extract specific tasks from the provided "Job Tasks" text in a recruitment poster. Please use the following instructions, recommended steps, samples and output format to respond to user inputs.

[User] Instructions:

- (1) For structure and content, format each task to be similar to O*NET tasks, such as "Direct or coordinate an organization's financial or budget activities to fund operations, maximize investments, or increase efficiency".
- (2) For length and language, each task that you extract is expected to be between 10-30 words in English.
- (3) Maintain consistency with the original text description as much as possible.
- (4) Some tasks in "Job Tasks" are long and complex such as "Operation, assembly, testing, inspection, packaging"; split it into small tasks which are consistent with the O*NET tasks.
- (5) Some tasks are simple and general such as "Product packaging"; integrate and summarize moderately in combination with the context.
- (6) Specific information such as place names, company names, and product names should not appear in the tasks you extract. Instead, generalized nouns and language should be used. For example, "Be responsible for the good credit of customers of [a named lender]" should be cleaned up as "Maintain good credit history of borrowers of the company's micro loan product".
- (7) The aim is to calculate the text similarity between these extracted tasks and O*NET tasks.

A two-step process is suggested: First, extract specific tasks from the provided "Job Tasks" text and organize the language and structure to make it comparable with O*NET tasks. Second, reread the given "Job Tasks" text and check the extracted job tasks to ensure that there are no duplications or omissions, and that the granularity and description of the tasks are consistent with those in O*NET.

Output format (Do not include anything else in the answer): {Task1 | Task2 | ...}.

For reference: <worked example on a Chinese-language posting for a course-sales role, yielding four task statements>

Here is the text from the recruitment poster: <job tasks and requirements>

Step 3: screen.

[System] Your task is to filter a "Job Tasks" list. Please use the following instructions, samples and output format to respond to user inputs.

[User]

- (1) Some task descriptions contain non-English characters, please return "This task is multilingual."
- (2) Some task descriptions are too broad, please return "This task is too broad." For example, the description "Complete other tasks assigned by the leader" is too general - we cannot determine the specific nature of the tasks from such a vague statement.
- (3) Some task descriptions are actually requirements, please return "This task is a requirement." For example, "Master foundational JAVA and J2EE technologies" outlines a requirement rather than a specific, actionable task.
- (4) Some task descriptions are overly detailed, including specific information like place names, company names, and product names. You should generalize such specifics and return the revised task.
- (5) For qualified task descriptions, simply return the original text.

The input is a set of tasks like "Task1 | Task2 | ...". The output format should be: "Processed Task1 | Processed Task2 | ...".

For reference, here is an example of correctly applied filters: <five-item example returning, in order, "This task is multilingual.", "This task is too broad.", "This task is a requirement.", and two generalized task statements>

Here is the text you need to filter: <extracted tasks>

F.4 Extended data and classification

The public sample extends coverage to January 2018–July 2025 (about 200,000 postings per year, with roughly 200,000 for the first seven months of 2025), and reports updated platform scale: about 374 million active individual users and 14.36 million firm users. The documentation is explicit that online postings over-represent urban, younger, better-educated, and higher-paying positions, so the index should be generalized to the labor market as a whole with care—the same caveat that applies to the estimates in the main text.

Because Zhaopin adopted a new internal occupational taxonomy in 2023, the team built a crosswalk between the platform's classification and the US SOC system to keep the series consistent over time. The crosswalk holds the series together, but the documentation notes that the change can still leave a break in 2023, most visibly in thin occupation groups and in small cities under the indirect city-level method.

F.5 Released data products

The release provides both base-period indices (averaged over 2018–2021) and dynamic annual indices, at several aggregation levels, as summarized in Table A4, and is updated annually. Each file reports the exposure index together with occupation identifiers and titles (in English and Chinese) and, for the dynamic files, the year. To keep cells reliably populated, each file is truncated to the units with the largest base-period sample shares: the top 60 SOC minor groups, and the top 100 SOC detailed occupations and Zhaopin second-level occupations.

Table A4: Public exposure-index data products (v2025.09 release)

Type	Aggregation level	Description
Base period	SOC minor group	Exposure over the 2018–2021 base period, by SOC minor group
Base period	SOC detailed occupation	Exposure over the 2018–2021 base period, by SOC detailed occupation
Base period	Zhaopin second-level occupation	Exposure over the 2018–2021 base period, by Zhaopin occupation
Dynamic (annual)	SOC detailed occupation	Year-by-year exposure, by SOC detailed occupation
Dynamic (annual)	Zhaopin second-level occupation	Year-by-year exposure, by Zhaopin occupation
Dynamic (annual)	City (direct method)	Year-by-year city-level exposure

Notes: Summarized from the release documentation (AI and Economics Laboratory, National School of Development, Peking University and Zhaopin, 2025). Occupation levels follow the US SOC system: SOC detailed occupations (the finest unit), SOC minor groups (the “occupation groups” of the main text), and the platform’s own second-level occupational classification.